DATA ANALYTICS INTERNSHIP PROGRAM

ELECTRONIC PRODUCT SALES ANALYSIS

AN INTERNSHIP PROJECT REPORT

Submitted by

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(Reg No: 110725102277)

in partial fulfilment for the award of the degree of

of

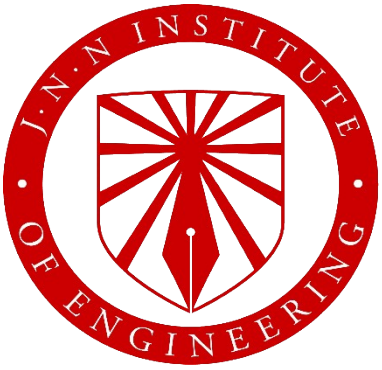
BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Internship carried out at

WIZTECH AUTOMATION SOLUTIONS, CHENNAI



J.N.N INSTITUTE OF ENGINEERING

(An Autonomous Institute, Affiliated to Anna University, Chennai)

AUGUST – 2026

VISION OF THE INSTITUTE

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instil in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

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VISION OF THE DEPARTMENT

To impart quality education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT

DM1: To enhance the knowledge of students with recent advancements in Artificial Intelligence and Data Science.

DM2: To equip students with strong fundamental concepts, analytical capability, programming and problem-solving skills.

DM3: To make students industry ready and enhance employability through training, internships and real-time projects.

DM4: To guide students to perform research on Artificial Intelligence and Data Science for solving industry problems.

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S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team.

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S.NO	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Foresight, Insight, hindsight from data for solving business and engineering problems.
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.

BONAFIDE CERTIFICATE

Certified that this internship project report titled **"ELECTRONIC PRODUCT SALES ANALYSIS"** is the bonafide work of **"R YATHIN YASHWANTH"** (**Reg No: 110725102277**), a student of the Department of Artificial Intelligence & Data Science, J.N.N Institute of Engineering, who completed the internship at Wiztech Automation Solutions, Chennai.

Signature of the HOD with date

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Signature of the Supervisor with date

Miss. M. Gayathridevi
Project Supervisor,
Department of Artificial Intelligence & Data Science,
J.N.N Institute of Engineering, Chennai

Certified that the above student has attended the viva voce examination held on .....

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.,** and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYACHANDRAN M.S.,** for providing the necessary facilities for the completion of my project.

I thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.,** for the encouragement provided with the congenial environment essential for me to complete the project.

I take this opportunity to express my gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving me continuous encouragement for completing my project.

I extend my sincere thanks to **Wiztech Automation Solutions, Chennai** and its **Director** for giving me the opportunity to undergo an internship in Data Analytics and for the guidance and support extended throughout the internship period from 05.08.2026 to 20.08.2026. I also thank the technical team of Wiztech Automation Solutions for their valuable support.

Finally, I would like to thank my parents and friends for their constant support and encouragement.

ABSTRACT

Electronic Product Sales Analysis is a data analytics project that studies the sales records of an electronic product company in order to understand which products, regions, sales channels and customer groups drive its business. The dataset used in this project contains **235 orders** described by **12 fields**, including order date, product name, unit price, quantity, discount rate, customer age, region, sales channel, payment method, sales representative and customer rating.

This report documents the Data Analytics internship completed by **R YATHIN YASHWANTH** at Wiztech Automation Solutions, Chennai, from 05.08.2026 to 20.08.2026. The project follows a structured workflow of data loading, data checking and cleaning, derived-measure calculation, exploratory analysis, statistical summaries and visualisation using Python, Pandas, NumPy, Matplotlib and Seaborn.

The analysis shows that the Tablet 10-Inch and Smartphone Pro Max are the most-sold products by units, that orders are spread almost evenly across four sales channels, and that customer ratings average 3.06 out of 5. The analysis also uncovered a data-quality issue: a single order with a unit price of $125,000 (against a normal price of $349 for the same tablet) accounts for more than half of the recorded revenue and makes the West region appear to dominate. The report quantifies this effect, presents results both as recorded and after excluding that record, and recommends verifying the entry at source.

The project gave practical exposure to data preparation, analysis, visualisation and interpretation, and connected classroom concepts with an industry-oriented analytics problem.

TABLE OF CONTENTS

CHAPTER NO	Title	Page
1	**INTRODUCTION**	**10**
	1.1 Overview	10
	1.2 Internship Details	10
	1.3 About the Organization	10
	1.4 Objectives	11
	1.5 Scope of the Project	11
	1.6 Internship Work Schedule	11
2	**LITERATURE REVIEW**	**12**
	2.1 Sales and Customer Analysis	12
	2.2 Data Analytics in Retail and Electronics	12
	2.3 Data Cleaning	12
	2.4 Exploratory Data Analysis	12
	2.5 Tools and Technologies	13
3	**METHODOLOGY**	**14**
	3.1 Overall Approach	14
	3.2 System Architecture	14
	3.3 Data Fields	15
	3.4 Analytical Modules	15
	3.5 Analysis Measures	15
4	**IMPLEMENTATION**	**16**
	4.1 Data Loading	16
	4.2 Data Cleaning	16
	4.3 Exploratory Analysis	17
	4.4 Customer Segmentation	17
	4.5 Visualisation	18
5	**RESULTS AND DISCUSSION**	**22**
	5.1 Results	22
	5.2 Discussion	26
	5.3 Challenges Faced	27
	5.4 Lessons Learned	27
6	**CONCLUSION AND FUTURE WORK**	**28**
	6.1 Conclusion	28
	6.2 Future Work	28
	6.3 Outcome Mapping	29
	REFERENCES	**30**
	APPENDICES	**31**
	INTERNSHIP COMPLETION CERTIFICATE	**39**

CHAPTER 1
INTRODUCTION

1.1 Overview

Data analytics is the process of collecting, cleaning, organising and examining data to discover useful patterns and support decision-making. For a company that sells consumer electronics, order-level sales data is one of the most valuable sources of information, because it records what was sold, at what price, in which region, through which channel, with what discount and how the customer rated the experience.

Electronic Product Sales Analysis applies these ideas to a sales dataset of 235 orders covering twelve products. The project focuses on checking and preparing the order records, calculating the revenue of every order, analysing sales by product, region, channel, payment method and customer age, and communicating the findings through tables and charts. The objective is to turn raw order records into information that business users can interpret and act on.

1.2 Internship Details

Particulars	Details
Student Name	R YATHIN YASHWANTH
Register Number	110725102277
Institution	J.N.N Institute of Engineering
Department	Artificial Intelligence & Data Science
Programme	B.Tech (AI & DS)
Organisation	Wiztech Automation Solutions, Chennai
Domain	Data Analytics
Project Title	Electronic Product Sales Analysis
Duration	05.08.2026 to 20.08.2026
Dataset	electronic_product_company_clean_data.csv (235 orders, 12 fields)

1.3 About the Organization

Wiztech Automation Solutions is located at New No. 13, 6th Street, A Block, near Anna Nagar East Roundtana, Anna Nagar, Chennai, Tamil Nadu 600102. The internship completion certificate issued by the organisation carries the ISO 9001:2008 and IAO accreditation marks. The internship provided exposure to the Data Analytics domain and project-based learning.

Particulars	Details
Organisation	Wiztech Automation Solutions
Location	Anna Nagar, Chennai, Tamil Nadu – 600102
Phone	044-2620 9369
Email	wiztech4automation@gmail.com
Website	wiztechautomationsolutions.com
Recognition	ISO 9001:2008; IAO accredited (as shown on the certificate)
Internship Domain	Data Analytics

1.4 Objectives

- Understand the fundamentals of sales and customer purchase analysis for an electronic product company.
- Check, clean and prepare the order records for analysis.
- Calculate order revenue and descriptive measures such as totals, averages, counts and shares.
- Identify best-selling products, strong and weak regions and channels, and customer patterns.
- Detect unusual records that can distort results and report their effect honestly.
- Use charts to communicate product-level, region-level and customer-level patterns.
- Gain practical exposure to Python, Pandas, NumPy, Matplotlib, Seaborn and SQL concepts.

1.5 Scope of the Project

The project covers descriptive analysis of the electronic product sales dataset. The scope includes data checking and preparation, exploratory analysis, product, region, channel, payment and customer summaries, revenue analysis and visualisation. Predictive modelling, recommendation systems, time-series forecasting and deployment of a production application are outside the basic scope and are listed as future enhancements. Time-based trend analysis is also left for future work because the Order_Date field is stored in several different text formats (see Section 4.2).

1.6 Internship Work Schedule

Period	Work Area
05.08.2026 – 07.08.2026	Orientation, understanding the project problem and the data analytics workflow
08.08.2026 – 11.08.2026	Understanding the dataset structure and fields
12.08.2026 – 14.08.2026	Data checking and cleaning
15.08.2026 – 17.08.2026	Exploratory analysis and revenue calculation
18.08.2026 – 19.08.2026	Product, region, channel and customer analysis; visualisation
20.08.2026	Final review, interpretation and documentation

CHAPTER 2
LITERATURE REVIEW

This chapter presents the basic concepts related to sales and customer analysis and the tools used in a typical data analytics workflow.

2.1 Sales and Customer Analysis

Sales analysis examines order records to understand commercial performance. Typical measures include number of orders, units sold, total revenue, average order value, discount levels and the contribution of each product, region or channel to the total. Customer analysis adds information such as age group, payment preference and satisfaction rating. Together these measures give a descriptive view of what is selling, where, to whom and how well customers feel about it.

2.2 Data Analytics in Retail and Electronics

Business analytics converts raw business data into information that supports operational and strategic decisions. In consumer electronics, where product prices range from tens to thousands of dollars and discounts are used frequently, order-level analysis helps managers compare products, judge whether discounts are being applied sensibly, and decide where to focus stock, promotion and sales effort. Grouping records by product, region or channel allows analysts to identify recurring patterns and compare performance across groups.

2.3 Data Cleaning

Data cleaning is an essential stage before analysis. Common issues include missing values, duplicate transactions, inconsistent text formats, incorrect data types, mixed date formats and invalid or extreme numeric values. Cleaning improves the reliability of the analysis and reduces errors in calculated results. Extreme values deserve particular care, because a single mistaken entry can dominate totals and averages.

2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the structure and distribution of data before drawing conclusions. It may include summary statistics, frequency tables, grouping, correlation analysis and visualisation. EDA helps identify patterns, unusual observations and relationships between variables.

2.5 Tools and Technologies

Tool / Technology	Purpose
Python	Programming and data analysis
Pandas	Data loading, cleaning, grouping and aggregation
NumPy	Numerical calculations and array operations
Matplotlib	Charts and plots
Seaborn	Statistical visualisation (correlation heatmap)
SQL / MySQL	Structured data storage and querying (equivalent queries in Appendix E)
Jupyter Notebook / VS Code	Development, analysis and testing
CSV	Source and exchange format of the dataset

CHAPTER 3
METHODOLOGY

3.1 Overall Approach

The project follows a standard data analytics lifecycle. The workflow begins with understanding the problem and the dataset, followed by data checking and preparation, derivation of revenue, exploratory analysis, aggregation, visualisation and interpretation.

Stage	Activity	Outcome
1	Requirement understanding	Defined analysis questions
2	Data collection	Electronic product sales dataset (CSV)
3	Data checking and cleaning	Consistent, analysis-ready data
4	Exploratory analysis	Summary statistics and patterns
5	Product / region / customer analysis	Grouped insights
6	Visualisation	Charts and tables
7	Testing and documentation	Verified analysis and report

3.2 System Architecture

The project is represented as a simple analytics pipeline: **Sales Data (CSV) → Cleaning (Pandas) → Analysis (Python) → Visualisation (Matplotlib / Seaborn) → Report.** The data layer contains the source order records. The processing layer performs cleaning, transformation and calculations. The presentation layer communicates the results through tables and charts.

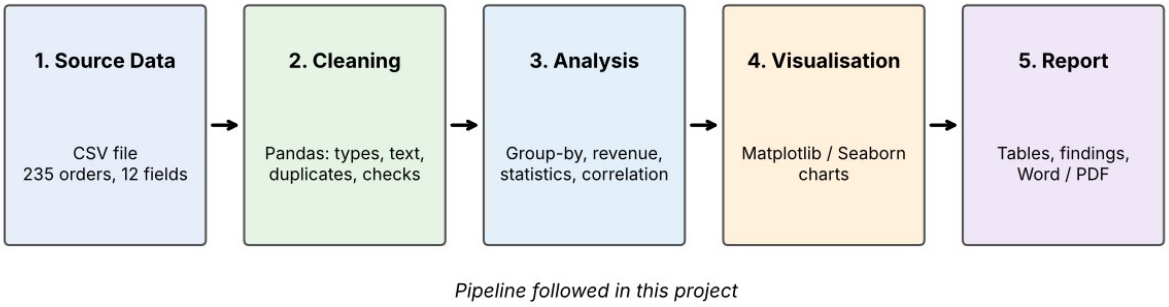


Fig. 1: System architecture of the analytics pipeline

3.3 Data Fields

Field	Type	Description
Order_ID	String	Unique identifier of an order (e.g., ORD-2025-1001)
Order_Date	String	Date of the order (stored in mixed text formats)
Product_Name	String	Name of the electronic product sold
Unit_Price	Float	Price of one unit in US dollars ($)
Quantity	Float	Number of units in the order (1 to 5)
Discount_Rate	Float	Discount applied, as a fraction (0.00 to 0.20)
Customer_Age	Float	Age of the customer in years
Region	String	Sales region (West, North, Central, East, South)
Sales_Channel	String	Channel of sale (Online Store, Retail Partner, Direct Sales, Wholesale)
Payment_Method	String	Method used to pay (Debit Card, Bank Transfer, etc.)
Sales_Rep	String	Sales representative who handled the order
Customer_Rating	Float	Customer satisfaction rating from 1 to 5

3.4 Analytical Modules

- **Data Preparation Module** – loads and validates the source dataset.
- **Cleaning Module** – checks missing values, duplicates, text formats, numeric types and unusual values.
- **Revenue Module** – derives the revenue of every order from price, quantity and discount.
- **Product Analysis Module** – identifies popular and high-value products.
- **Region and Channel Module** – compares revenue and orders across regions, channels and payment methods.
- **Customer Analysis Module** – summarises customers by age group, order value and rating.
- **Visualisation Module** – converts analytical results into understandable charts.

3.5 Analysis Measures

- **Order Revenue** = Unit_Price × Quantity × (1 − Discount_Rate).
- **Total Revenue** = Sum of the revenue of all valid orders.
- **Average Order Value** = Total revenue divided by the number of orders.
- **Units Sold** = Total quantity sold for a product.
- **Share (%)** = Group value compared with the overall value.
- **Average Discount** = Mean of Discount_Rate for a group.
- **Average Rating** = Mean of Customer_Rating for a group.

CHAPTER 4
IMPLEMENTATION

4.1 Data Loading

The sales dataset was loaded into Python using Pandas. The file contains **235 rows and 12 columns**: five numeric fields (Unit_Price, Quantity, Discount_Rate, Customer_Age, Customer_Rating) and seven text fields. All 12 columns are fully populated.

```
import pandas as pd

df = pd.read_csv("electronic_product_company_clean_data.csv")
print(df.head())
print(df.info())
print(df.isna().sum().sum())          # 0 missing values
print(df["Order_ID"].duplicated().sum())  # 0 duplicates
```

4.2 Data Cleaning

The cleaning stage checks for missing values, duplicate records, inconsistent text and invalid numeric values. The file supplied for the project is already a cleaned version, and the checks confirmed the following:

- **Missing values:** none – every column has 235 non-null values.
- **Duplicates:** none – all 235 Order_ID values are unique.
- **Text fields:** product names, regions, channels, payment methods and sales representatives are consistently formatted.
- **Customer_Age:** 7 records have an age of exactly 43.5, which is the median age of the dataset; this is consistent with missing ages having been filled with the median during earlier cleaning.
- **Order_Date:** stored as text in several formats (for example 2025-11-24, 07/03/2025, 2025/11/18, "Mar 13, 2025" and 29/09/2025). Some values such as 07/03/2025 can be read as day/month or month/day, so a reliable calendar could not be built. Date-based trend analysis was therefore not performed and is proposed as future work.
- **Extreme value:** order ORD-2025-1059 records a Tablet 10-Inch with a unit price of $125,000, whereas every other Tablet order is priced at $349. This is almost certainly a data-entry error (see Section 5.1).

```
df = df.drop_duplicates(subset="Order_ID")
for col in
["Product_Name","Region","Sales_Channel","Payment_Method","Sales_Rep"]:
    df[col] = df[col].astype(str).str.strip()
for col in
["Unit_Price","Quantity","Discount_Rate","Customer_Age","Customer_Rating"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")
df = df.dropna(subset=["Order_ID","Unit_Price","Quantity"])
df["Revenue"] = df["Unit_Price"] * df["Quantity"] * (1 - df["Discount_Rate"])
```

4.3 Exploratory Analysis

Exploratory analysis was performed using descriptive statistics and grouped summaries. Product-level grouping shows popular and high-value products, region and channel grouping shows where sales come from, and customer-level grouping shows how age and order value relate to ratings.

```
print(df[["Unit_Price","Quantity","Discount_Rate",
          "Customer_Age","Customer_Rating","Revenue"]].describe())

product_summary = df.groupby("Product_Name").agg(
    orders=("Order_ID","count"),
    units=("Quantity","sum"),
    revenue=("Revenue","sum")).sort_values("units", ascending=False)

region_revenue = df.groupby("Region")
["Revenue"].sum().sort_values(ascending=False)
```

The descriptive statistics of all 235 recorded orders are shown below. The very large maximum and standard deviation of Unit_Price and Revenue come from the single extreme record described above.

Field	Count	Mean	Std Dev	Min	Median	Max	75%
Unit_Price ($)	235	1,140.72	8,128.60	49.99	499.00	125,000.00	1,099.00
Quantity	235	3.24	1.37	1.00	3.00	5.00	4.00
Discount_Rate	235	0.09	0.07	0.00	0.10	0.20	0.15
Customer_Age	235	43.16	13.51	18.00	43.50	68.00	55.00
Customer_Rating	235	3.06	1.43	1.00	3.00	5.00	4.00
Revenue ($)	235	4,014.77	34,579.04	39.99	1,182.75	531,250.00	2,697.60

4.4 Customer Segmentation

A simple descriptive segmentation was applied. The dataset does not contain a customer identifier, so customers are grouped by **age group** and orders are grouped into **value tiers** (Low, Medium, High) using the lower and upper thirds of order revenue. These categories are analytical groupings rather than fixed business labels. Order-value tiers and age-group revenue exclude the extreme record ORD-2025-1059 so that it does not distort the groups.

```
df["Age_Group"] = pd.cut(df["Customer_Age"],
    bins=[17,29,39,49,59,70],
    labels=["18-29","30-39","40-49","50-59","60-68"])

ok = df[df["Unit_Price"] < 5000]           # exclude ORD-2025-1059
ok["Value_Tier"] = pd.qcut(ok["Revenue"], q=3,
    labels=["Low","Medium","High"])
```

Table 4.1: Customer segments by age group

Age group	Orders	Avg order value ($)	Total revenue ($)	Avg rating
18-29	52	1,852.68	96,339.56	3.15
30-39	40	1,553.11	62,124.36	3.27
40-49	62	1,862.63	113,620.54	2.79
50-59	47	1,359.73	63,907.47	2.98
60-68	34	2,242.04	76,229.44	3.26

Table 4.2: Order-value tiers (tier boundaries: Low up to $660.47, Medium up to $2,037.45, High above)

Tier	Orders	Revenue ($)	Revenue share	Avg discount	Avg rating
Low	78	24,623.18	6.0%	11.0%	3.01
Medium	79	98,734.15	24.0%	8.5%	3.19
High	77	288,864.05	70.1%	8.6%	3.00

The High tier — only about one order in three — generates **70.1%** of revenue, while the Low tier contributes only 6.0%. Low-value orders also carry the highest average discount (11.0%), which suggests that discounts are most heavily used on small orders.

4.5 Visualisation

Visualisation makes the results easier to interpret. Five charts were produced with Matplotlib: a scatter plot of average unit price against average discount rate, a bar chart of revenue by region, a bar chart of the top ten products by units sold, a pie chart of orders by sales channel and a histogram of customer age. The complete script is given in Appendix F.

```
import matplotlib.pyplot as plt

df.groupby("Region")["Revenue"].sum().sort_values(ascending=False) \
  .plot(kind="bar", color="steelblue", title="Total Revenue by Region")
plt.xlabel("Region"); plt.ylabel("Revenue ($)")
plt.xticks(rotation=45)
plt.show()
```

The charts are presented below and interpreted in Chapter 5. In Fig. 2 and Fig. 3, note that the Tablet 10-Inch average price (about $4,504) and the West region revenue (about $611,000) are inflated by the single extreme record ORD-2025-1059; the corrected view is given in Fig. 7.

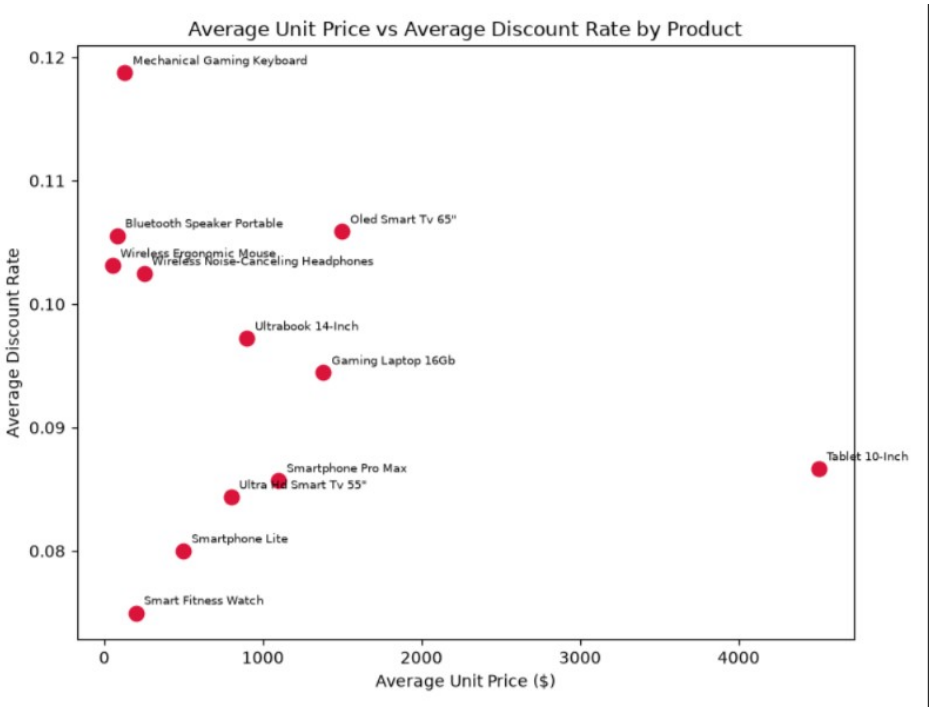


Fig. 2: Average Unit Price vs Average Discount Rate by Product

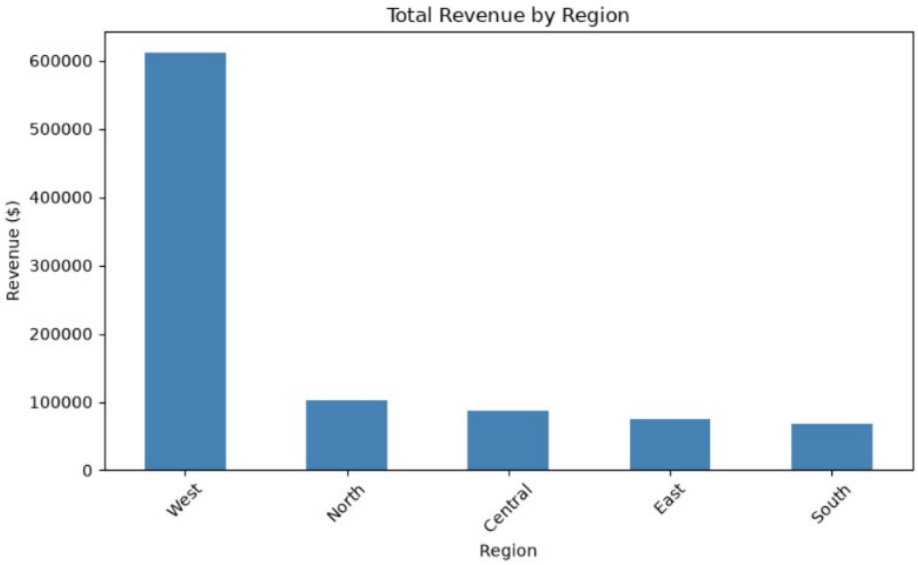


Fig. 3: Total Revenue by Region (as recorded)

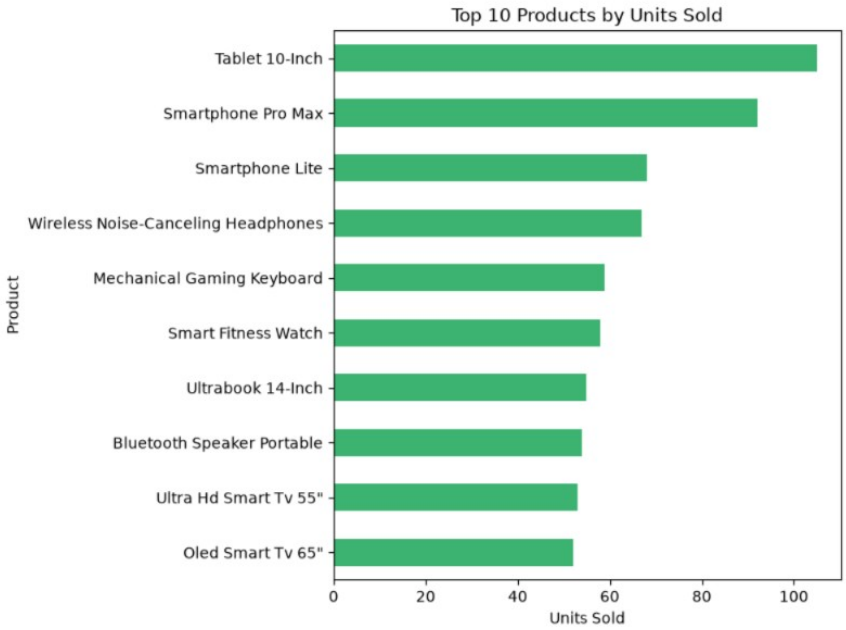


Fig. 4: Top 10 Products by Units Sold

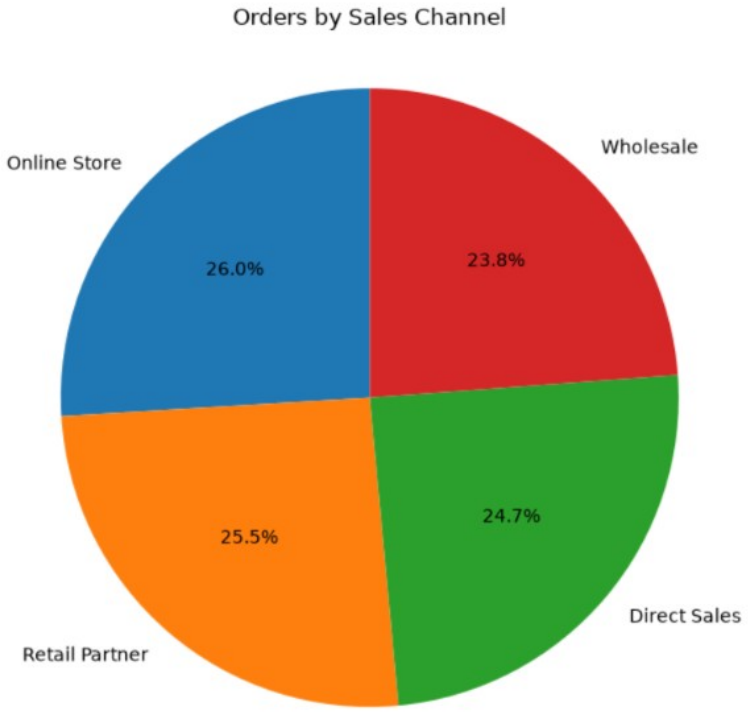


Fig. 5: Orders by Sales Channel

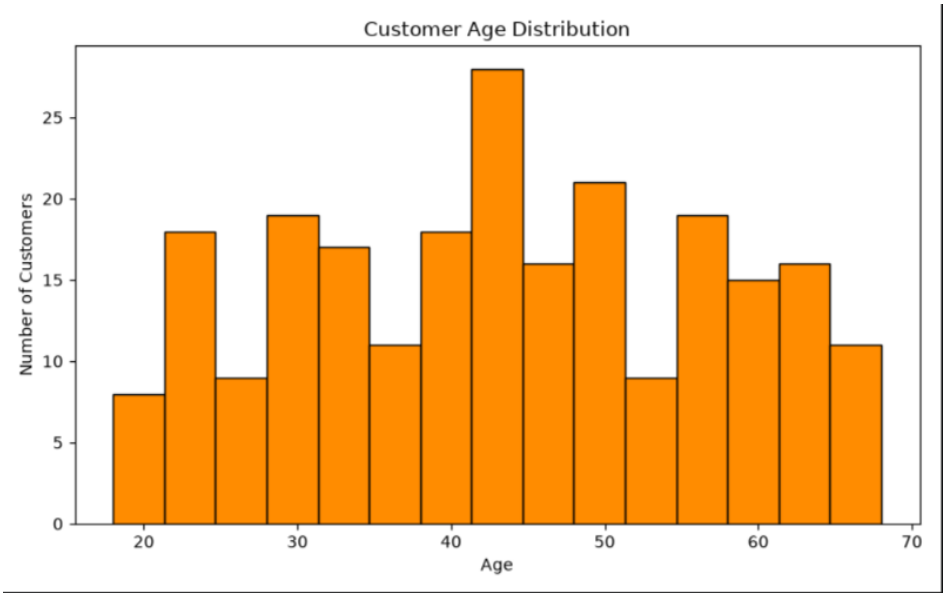


Fig. 6: Customer Age Distribution

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Results

The analysis produced a structured summary of the company's sales. Unless stated otherwise, figures are calculated from all 235 recorded orders; where one extreme record would distort a result, figures are also given after excluding it.

5.1.1 Overall sales

Measure	Value
Orders analysed	235
Units sold	762
Total revenue as recorded	$943,471.38
Total revenue excluding ORD-2025-1059	$412,221.38
Revenue of ORD-2025-1059 alone	$531,250.00 (56.3% of recorded revenue)
Average order value (excluding ORD-2025-1059)	$1,761.63
Median order value	$1,182.75
Average discount rate	9.4%
Average customer rating	3.06 / 5

5.1.2 Product analysis

Fig. 4 shows that the **Tablet 10-Inch (105 units)** and the **Smartphone Pro Max (92 units)** are the two most-sold products, followed by the Smartphone Lite (68) and the Wireless Noise-Canceling Headphones (67). The remaining products each sold between 48 and 59 units, so unit sales are fairly evenly spread across the range. Revenue is led by the Smartphone Pro Max, the Oled Smart Tv 65" and the Gaming Laptop, which have high unit prices.

Table 5.1: Product summary (revenue excludes ORD-2025-1059; price is the median unit price)

Product	Orders	Units	Price ($)	Avg disc.	Rating	Revenue ($)
Tablet 10-Inch	30	105	349.00	8.7%	3.23	32,195
Smartphone Pro Max	28	92	1,099.00	8.6%	3.18	92,316
Smartphone Lite	20	68	499.00	8.0%	3.20	31,038
Wireless Noise-Canceling Headphones	20	67	249.00	10.3%	3.05	15,152
Mechanical Gaming Keyboard	16	59	129.00	11.9%	2.69	6,663
Smart Fitness Watch	18	58	199.00	7.5%	3.17	10,716
Ultrabook 14-Inch	18	55	899.00	9.7%	3.17	44,635
Bluetooth Speaker Portable	18	54	79.99	10.6%	2.67	3,884
Ultra Hd Smart Tv 55"	16	53	799.00	8.4%	3.19	38,632
Oled Smart Tv 65"	17	52	1,499.00	10.6%	2.71	69,629
Gaming Laptop 16Gb	18	51	1,399.00	9.4%	2.83	65,203
Wireless Ergonomic Mouse	16	48	49.99	10.3%	3.44	2,160

Fig. 2 shows that the **Mechanical Gaming Keyboard** has the highest average discount (11.9%) and the **Smart Fitness Watch** the lowest (7.5%). Low-priced accessories tend to receive higher discounts, while mid- and high-priced items such as the Smartphone Lite and the Ultra Hd Smart Tv 55" receive lower ones. Customer ratings are highest for the Wireless Ergonomic Mouse (3.44) and lowest for the Bluetooth Speaker Portable (2.67), the Mechanical Gaming Keyboard (2.69) and the Oled Smart Tv 65" (2.71).

5.1.3 Region analysis

As recorded, the **West region** shows revenue of about $611,000 — more than five times any other region (Fig. 3). This is misleading: the West total includes the extreme order ORD-2025-1059 (a Tablet at an entered price of $125,000, revenue $531,250). Excluding that single record, West revenue falls to about $80,000 and the ranking changes completely, as shown in Fig. 7 and Table 5.2.

Table 5.2: Revenue by region

Region	Orders	Revenue as recorded ($)	Revenue excl. ORD-2025-1059 ($)	Avg rating
West	43	611,342.94	80,092.94	3.00
North	50	102,932.04	102,932.04	2.76
Central	47	86,612.19	86,612.19	3.15
East	53	75,184.37	75,184.37	3.26
South	42	67,399.84	67,399.84	3.12

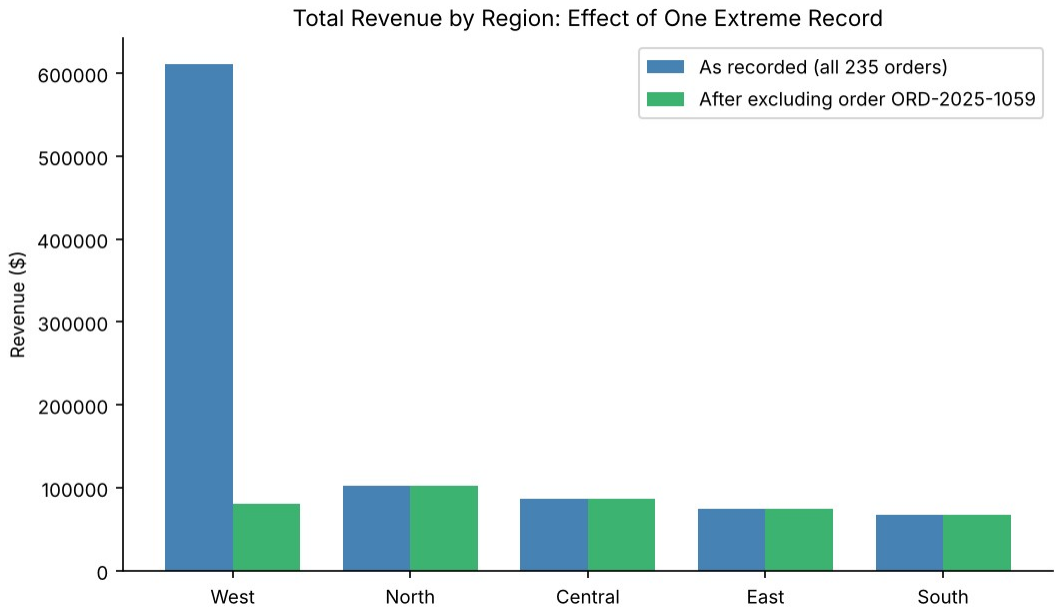


Fig. 7: Effect of one extreme record on regional revenue

After correction, **North** leads with about $102,900, followed by Central ($86,600), West ($80,100), East ($75,200) and South ($67,400). Revenue is therefore spread much more evenly across regions than the uncorrected chart suggests.

5.1.4 Sales channel and payment analysis

Orders are almost evenly divided among the four channels (Fig. 5): Online Store 26.0%, Retail Partner 25.5%, Direct Sales 24.7% and Wholesale 23.8%. By revenue (excluding the extreme record), **Retail Partner** is the strongest channel at about $116,300. Wholesale customers give the highest average rating (3.16). The Online Store and Wholesale channels use the highest average discounts (about 10.1%), while Direct Sales uses the lowest (8.1%).

Table 5.3: Sales channel summary (revenue excludes ORD-2025-1059)

Channel	Orders	Share	Revenue ($)	Avg disc.	Rating
Online Store	61	26.0%	97,122.53	10.1%	3.02
Retail Partner	60	25.5%	116,275.10	9.3%	3.03
Direct Sales	58	24.7%	96,170.33	8.1%	3.03
Wholesale	56	23.8%	102,653.42	10.1%	3.16

Debit Card is the most common payment method (48 orders, 20.4%), followed by Bank Transfer (42) and Cash On Delivery (41). Credit Card is the least used (30 orders, 12.8%). Digital and bank-based methods together account for the large majority of orders (Fig. 8).

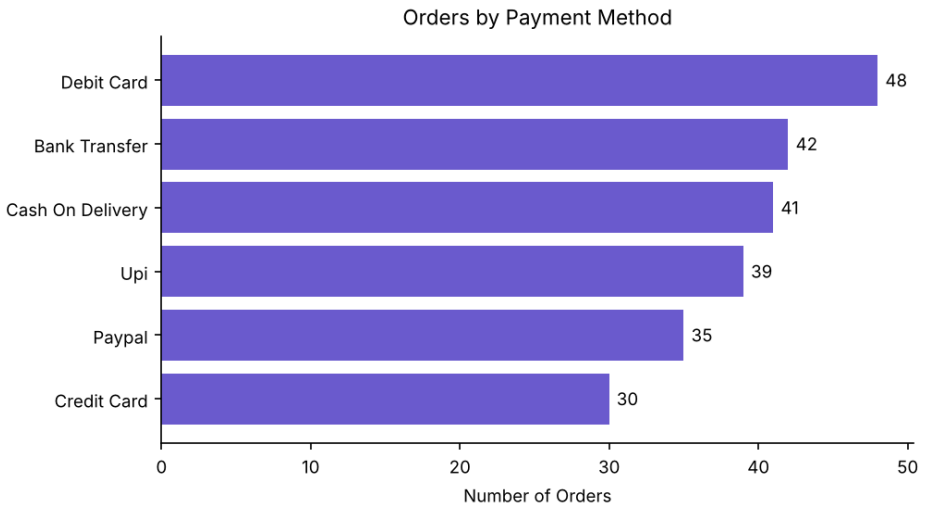


Fig. 8: Orders by Payment Method

5.1.5 Customer analysis

Customer ages range from 18 to 68 with a mean of 43.2 and a median of 43.5 years (Fig. 6). The age distribution is broad rather than concentrated: the 40–49 group is the largest (62 orders), followed by 18–29 (52) and 50–59 (47), while the 60–68 group is the smallest (34 orders) but has the highest average order value ($2,242). Customers aged 40–49 give the lowest average rating (2.79).

Customer ratings are spread almost evenly over the five levels (1★: 45, 2★: 46, 3★: 47, 4★: 44, 5★: 53 orders), giving an average of 3.06. About 38.7% of orders were rated 1 or 2, so a sizeable share of customers were dissatisfied. Discount rates take only five values — 0%, 5%, 10%, 15% and 20% — and are used on 76.6% of orders.

5.1.6 Sales representative performance

Sales representative	Orders	Revenue excl. ORD-2025-1059 ($)	Avg rating
Liam O'Connor	39	82,390.24	3.51
Sofia Garcia	32	68,267.62	3.25
Priya Patel	37	67,008.10	2.68
David Kim	30	53,763.86	3.03
Sarah Jenkins	27	40,332.19	3.07
Michael Chen	27	38,351.28	2.93
Emma Wilson	23	34,330.82	2.91
Alex Rivera	20	27,777.28	2.95

Liam O'Connor handled the most orders (39) and has the highest revenue (about $82,400) and the highest average rating (3.51). Priya Patel handled 37 orders; her recorded revenue of about $598,000 is inflated by ORD-2025-1059, and after exclusion it is about $67,000, with the lowest average rating (2.68).

5.2 Discussion

Electronic Product Sales Analysis demonstrates how raw order records can be converted into understandable business information. Grouping and aggregation make it possible to compare products, regions and channels, while visualisation makes larger patterns easier to observe. The main points of discussion are:

- **Data quality decides the conclusion.** One mistaken price changed the apparent leading region from North to West and made the tablet look like a $4,500 product. Always check for extreme values before trusting totals and averages.
- **Sales are broad-based.** Unit sales are spread over twelve products and orders over four channels, so the business is not dependent on a single product or channel.
- **Revenue depends on price, not discount.** In the corrected data, revenue correlates strongly with unit price ($r = 0.80$) and moderately with quantity ($r = 0.40$), while discount rate has a weak negative relationship ($r = -0.14$) and no useful relationship with customer rating ($r = -0.08$) (Appendix C). Heavier discounts did not buy higher satisfaction.
- **Discounts are weighted towards low-value orders.** The Low value tier has the highest average discount (11.0%) and the smallest revenue share (6.0%); discount policy for small orders can be reviewed.
- **Customer satisfaction needs attention.** The average rating is only 3.06 out of 5, and products such as the Bluetooth Speaker, Mechanical Gaming Keyboard and Oled Smart Tv 65" rate below 2.8.

The quality of the conclusions depends on the quality, completeness and representativeness of the dataset. The dataset covers 235 orders, so the findings describe this sample and should be validated on larger data before being used for firm business decisions.

5.3 Challenges Faced

- Understanding the structure and meaning of each dataset field.
- Dealing with an extreme price entry that distorted totals, averages and charts.
- Handling the Order_Date column, which mixes several text date formats and ambiguous day/month orders.
- Selecting appropriate grouping and aggregation methods.
- Choosing charts that communicate the intended result clearly.
- Interpreting analytical outputs without making unsupported conclusions.

5.4 Lessons Learned

- Data cleaning and validation are the foundation of reliable analysis.
- A single outlier can reverse a conclusion, so results should be checked with and without extreme values.
- Pandas grouping and aggregation are powerful for product, region and channel summaries.
- Visualisation improves the communication of analytical findings.
- SQL and Python can be used together for structured data analysis.
- Documentation and validation are important parts of an analytics project.
- Internship work helps connect academic concepts with practical data problems.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion

The Electronic Product Sales Analysis project provided practical exposure to the data analytics workflow during the internship at Wiztech Automation Solutions, Chennai. The project focused on preparing the order data, deriving revenue, performing descriptive analysis and presenting sales patterns through suitable visualisations.

The analysis found that the Tablet 10-Inch and Smartphone Pro Max lead unit sales, that orders are evenly spread across the four channels, that revenue is driven mainly by unit price, and that the average customer rating is a modest 3.06 out of 5. It also found that a single entry error (ORD-2025-1059) accounts for 56.3% of the recorded revenue and must be corrected at source before regional or product revenue figures are used for decisions.

The internship strengthened practical understanding of Python-based data analysis, data cleaning, grouping, statistical summaries and visual communication. It also provided an opportunity to apply concepts learned in the Artificial Intelligence & Data Science curriculum to an industry-oriented problem.

6.2 Future Work

- Correct or verify the extreme price entry and re-run all revenue results.
- Standardise the Order_Date field and add monthly and seasonal trend analysis.
- Develop a predictive model for future sales and customer ratings.
- Build product recommendation systems based on historical orders.
- Use clustering techniques for advanced customer segmentation.
- Create an interactive Power BI or Tableau dashboard and integrate the analysis with a web-based application.
- Add automated data pipelines and validation rules for regular analysis.

6.3 Outcome Mapping

Outcome	How the Internship Contributed
PO1 – Engineering Knowledge	Applied programming, statistics and data analytics concepts.
PO2 – Problem Analysis	Converted a business problem into measurable analytical tasks and detected a data-quality problem.
PO3 – Design/Development	Designed a structured workflow for data preparation and analysis.
PO5 – Modern Tool Usage	Used Python, Pandas, NumPy, Matplotlib, Seaborn and SQL concepts.
PO9 – Individual and Team Work	Developed professional communication and task-management skills.
PO10 – Communication	Presented analytical findings through tables, charts and documentation.
PO12 – Life-long Learning	Gained practical exposure to industry-oriented data analytics.

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APPENDIX A
DATASET DESCRIPTION

This appendix summarises the dataset used in the project, electronic_product_company_clean_data.csv, which contains **235 entries and 12 columns**.

A.1 Column Structure

No.	Column	Non-Null Count	Dtype	Description
0	Order_ID	235	object	Unique order code
1	Order_Date	235	object	Order date (mixed text formats)
2	Product_Name	235	object	Product sold (12 distinct products)
3	Unit_Price	235	float64	Price per unit in dollars
4	Quantity	235	float64	Units ordered (1–5)
5	Discount_Rate	235	float64	Discount fraction (0–0.20)
6	Customer_Age	235	float64	Customer age in years (18–68)
7	Region	235	object	West, North, Central, East, South
8	Sales_Channel	235	object	4 channels
9	Payment_Method	235	object	6 payment methods
10	Sales_Rep	235	object	8 sales representatives
11	Customer_Rating	235	float64	Rating from 1 to 5

A.2 Sample Records

Order_ID	Date	Product	Price	Qty	Disc.	Age	Region
ORD-2025-1001	2025-11-24	Smartphone Pro Max	1099.0	3	0.05	32	South
ORD-2025-1002	2025-01-16	Oled Smart Tv 65"	1499.0	2	0.20	56	North
ORD-2025-1003	2025-05-23	Ultra Hd Smart Tv 55"	799.0	4	0.10	35	South
ORD-2025-1004	07/03/2025	Tablet 10-Inch	349.0	3	0.00	64	West

A.3 Observations on Data Quality

- All 12 columns show 235 non-null values, so the file contains **no missing entries**; all Order_ID values are unique.
- **Order_Date** is stored as text in mixed formats and must be standardised before any time-based analysis.
- **Customer_Age** contains 7 values of exactly 43.5 (the median), consistent with median filling of missing ages.
- **Unit_Price** contains one extreme value ($125,000 for a Tablet 10-Inch, order ORD-2025-1059) against a normal tablet price of $349.

APPENDIX B
INTERPRETATION OF DESCRIPTIVE STATISTICS

This appendix restates the descriptive statistics for the numeric columns and adds a few derived measures. Table B.1 uses all recorded orders; Table B.2 excludes order ORD-2025-1059.

B.1 Summary Statistics (all 235 orders)

Statistic	Unit_Price ($)	Quantity	Customer_Age	Rating	Revenue ($)
Count	235.00	235.00	235.00	235.00	235.00
Mean	1,140.72	3.24	43.16	3.06	4,014.77
Std deviation	8,128.60	1.37	13.51	1.43	34,579.04
Minimum	49.99	1.00	18.00	1.00	39.99
25th percentile	199.00	2.00	32.50	2.00	438.60
Median	499.00	3.00	43.50	3.00	1,182.75
75th percentile	1,099.00	4.00	55.00	4.00	2,697.60
Maximum	125,000.00	5.00	68.00	5.00	531,250.00

B.2 Order Revenue excluding ORD-2025-1059

Measure	Value
Mean	$1,761.63
Standard deviation	$1,646.01
25th percentile (Q1)	$438.60
Median	$1,182.75
75th percentile (Q3)	$2,697.00
Maximum	$7,495.00
Inter-quartile range (Q3 − Q1)	$2,258.40
Upper outlier fence (Q3 + 1.5 × IQR)	$6,084.60
Mean − median gap	$578.88 (mean is 48.9% above the median)

B.3 Interpretation

- **Right-skewed order values.** The mean ($1,761.63) is well above the median ($1,182.75), so a smaller number of high-value orders pull the average upwards. The median is a more representative figure for a typical order.
- **High-value orders.** Orders above the Tukey fence of $6,084.60 are unusually large for the dataset and can be reviewed separately.
- **The extreme record.** In the full data, Unit_Price has a mean of $1,140.72 against a median of $499.00 and a standard deviation of $8,128.60 — all caused by the $125,000 entry.

APPENDIX C
CORRELATION ANALYSIS

The correlation heatmap shows pairwise Pearson correlation coefficients between the numeric columns, calculated after excluding order ORD-2025-1059. A value close to +1 or −1 indicates a strong linear relationship, while a value close to 0 indicates little or no linear relationship.

C.1 Correlation Matrix

	Price	Quantity	Discount	Age	Rating	Revenue
Unit_Price	1.00	-0.06	-0.02	0.04	-0.04	0.80
Quantity	-0.06	1.00	-0.08	-0.05	-0.01	0.40
Discount_Rate	-0.02	-0.08	1.00	0.14	-0.08	-0.14
Customer_Age	0.04	-0.05	0.14	1.00	-0.04	0.03
Customer_Rating	-0.04	-0.01	-0.08	-0.04	1.00	-0.04
Revenue	0.80	0.40	-0.14	0.03	-0.04	1.00

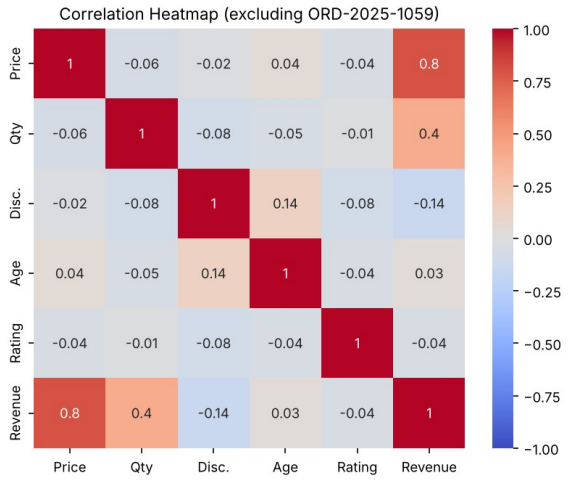


Fig. 9: Correlation heatmap of numeric fields

C.2 Interpretation

- **Unit_Price and Revenue (r = 0.80).** A strong positive relationship: order revenue is driven mainly by the price of the product.
- **Quantity and Revenue (r = 0.40).** A moderate positive relationship: larger orders add revenue, but less than price does.
- **Discount_Rate and Revenue (r = −0.14).** A weak negative relationship: higher discounts slightly reduce order revenue.
- **Customer_Rating (|r| ≤ 0.09 with every other field).** Rating is essentially unrelated to price, quantity, discount and age, so none of these explains customer satisfaction.

APPENDIX D
PRODUCT AND SEGMENT ANALYSIS

Fig. 4 ranks the ten best-selling products by units. The table below restates the units for all twelve products and adds each product's share of total units sold (762 units).

Rank	Product	Units sold	Share of units	Cumulative share
1	Tablet 10-Inch	105	13.8%	13.8%
2	Smartphone Pro Max	92	12.1%	25.9%
3	Smartphone Lite	68	8.9%	34.8%
4	Wireless Noise-Canceling Headphones	67	8.8%	43.6%
5	Mechanical Gaming Keyboard	59	7.7%	51.3%
6	Smart Fitness Watch	58	7.6%	58.9%
7	Ultrabook 14-Inch	55	7.2%	66.1%
8	Bluetooth Speaker Portable	54	7.1%	73.2%
9	Ultra Hd Smart Tv 55"	53	7.0%	80.2%
10	Oled Smart Tv 65"	52	6.8%	87.0%
11	Gaming Laptop 16Gb	51	6.7%	93.7%
12	Wireless Ergonomic Mouse	48	6.3%	100.0%

D.1 Observations

- **Tablet 10-Inch leads** with 105 units (13.8% of all units), followed by Smartphone Pro Max (92 units).
- The top four products together account for 43.6% of units; the remaining eight products each contribute between 6% and 8%.
- By revenue (excluding ORD-2025-1059), **Smartphone Pro Max ($92,316)**, **Oled Smart Tv 65" ($69,629)** and **Gaming Laptop 16Gb ($65,203)** lead.
- Low-priced accessories such as the Wireless Ergonomic Mouse and Bluetooth Speaker sell in similar unit numbers to premium items but contribute very little revenue.
- Higher-ranking products can be prioritised in stock planning and promotions, while lower-revenue accessories can be studied for bundling or targeted offers.

APPENDIX E
SQL QUERIES FOR STRUCTURED ANALYSIS

The following MySQL queries are equivalent to the main analysis measures defined in Section 3.5 and use the column names of the dataset. They are provided as a reference for structured storage and querying.

E.1 Table Definition

```
CREATE TABLE electronic_sales (
    order_id        VARCHAR(20),
    order_date      DATE,
    product_name    VARCHAR(60),
    unit_price      DECIMAL(10,2),
    quantity        INT,
    discount_rate   DECIMAL(4,2),
    customer_age    DECIMAL(4,1),
    region          VARCHAR(20),
    sales_channel   VARCHAR(30),
    payment_method  VARCHAR(30),
    sales_rep       VARCHAR(40),
    customer_rating INT
);
```

E.2 Total Revenue and Average Order Value

```
SELECT COUNT(*)                                      AS orders,
       SUM(unit_price*quantity*(1-discount_rate))    AS total_revenue,
       AVG(unit_price*quantity*(1-discount_rate))    AS avg_order_value
FROM   electronic_sales;
```

E.3 Revenue by Region

```
SELECT region,
       SUM(unit_price*quantity*(1-discount_rate)) AS revenue
FROM   electronic_sales
GROUP  BY region
ORDER  BY revenue DESC;
```

E.4 Top Products by Units Sold

```
SELECT product_name, SUM(quantity) AS units_sold
FROM   electronic_sales
GROUP  BY product_name
ORDER  BY units_sold DESC
LIMIT  10;
```

E.5 Orders by Sales Channel

```
SELECT sales_channel,
       COUNT(*) AS orders,
       ROUND(100*COUNT(*)/(SELECT COUNT(*) FROM electronic_sales),1) AS share_pct
FROM   electronic_sales
GROUP  BY sales_channel
ORDER  BY orders DESC;
```

APPENDIX F
COMPLETE PYTHON SCRIPT

The script below combines the loading, cleaning, revenue calculation, analysis and visualisation steps of Chapter 4 into one program. Running it on the dataset reproduces the five charts shown in Section 4.5 (Fig. 2 to Fig. 6).

```
import pandas as pd
import matplotlib.pyplot as plt

CSV = "electronic_product_company_clean_data.csv"
df = pd.read_csv(CSV)

# ---- 1. Basic checks / cleaning -----------------------------------
df = df.drop_duplicates(subset="Order_ID")
for col in ["Product_Name", "Region", "Sales_Channel",
            "Payment_Method", "Sales_Rep"]:
    df[col] = df[col].astype(str).str.strip()
for col in ["Unit_Price", "Quantity", "Discount_Rate",
            "Customer_Age", "Customer_Rating"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")
df = df.dropna(subset=["Order_ID", "Unit_Price", "Quantity"])

# ---- 2. Derived column --------------------------------------------
df["Revenue"] = df["Unit_Price"] * df["Quantity"] * (1 - df["Discount_Rate"])

# ---- 3. Descriptive summaries -------------------------------------
print(df[["Unit_Price", "Quantity", "Discount_Rate",
          "Customer_Age", "Customer_Rating", "Revenue"]].describe())
print(df.groupby("Region")["Revenue"].sum().sort_values(ascending=False))

# ---- 4. Charts ----------------------------------------------------
# 4.1 Average unit price vs average discount rate by product
prod = df.groupby("Product_Name").agg(price=("Unit_Price", "mean"),
                                      disc=("Discount_Rate", "mean"))
plt.figure(figsize=(8, 6))
plt.scatter(prod["price"], prod["disc"], s=90, color="crimson")
for name, r in prod.iterrows():
    plt.annotate(name, (r["price"], r["disc"]), fontsize=7,
                 xytext=(6, 5), textcoords="offset points")
plt.title("Average Unit Price vs Average Discount Rate by Product")
plt.xlabel("Average Unit Price ($)")
plt.ylabel("Average Discount Rate")
plt.savefig("out/1_price_vs_discount.png", dpi=120, bbox_inches="tight")
plt.close()

# 4.2 Total revenue by region
df.groupby("Region")["Revenue"].sum().sort_values(ascending=False) \
  .plot(kind="bar", color="steelblue", figsize=(8, 5),
        title="Total Revenue by Region")
plt.xlabel("Region"); plt.ylabel("Revenue ($)")
plt.xticks(rotation=45)
plt.savefig("out/2_revenue_by_region.png", dpi=120, bbox_inches="tight")
plt.close()

# 4.3 Top 10 products by units sold
df.groupby("Product_Name")["Quantity"].sum().sort_values().tail(10) \
  .plot(kind="barh", color="mediumseagreen", figsize=(7, 6),
        title="Top 10 Products by Units Sold")
plt.xlabel("Units Sold"); plt.ylabel("Product")
plt.savefig("out/3_top_products.png", dpi=120, bbox_inches="tight")
plt.close()

# 4.4 Orders by sales channel
```

```
df["Sales_Channel"].value_counts().plot(
    kind="pie", autopct="%1.1f%%", startangle=90, figsize=(6, 6),
    title="Orders by Sales Channel")
plt.ylabel("")
plt.savefig("out/4_orders_by_channel.png", dpi=120, bbox_inches="tight")
plt.close()

# 4.5 Customer age distribution
plt.figure(figsize=(8, 5))
plt.hist(df["Customer_Age"], bins=15, color="darkorange", edgecolor="black")
plt.title("Customer Age Distribution")
plt.xlabel("Age"); plt.ylabel("Number of Customers")
plt.savefig("out/5_age_distribution.png", dpi=120, bbox_inches="tight")
plt.close()
print("5 charts saved to out/")
```

APPENDIX G
KEY FINDINGS, WORK SUMMARY AND GLOSSARY

G.1 Key Findings

- The dataset contains 235 orders with no missing values across its 12 columns.
- Tablet 10-Inch (105 units) and Smartphone Pro Max (92 units) are the best-selling products.
- One order (ORD-2025-1059, Tablet at $125,000 per unit) contributes $531,250 — 56.3% of recorded revenue — and is very likely a data-entry error.
- Excluding that order, total revenue is $412,221.38 and North is the leading region ($102,932).
- Orders are split almost equally across four sales channels; Retail Partner earns the most revenue.
- Revenue correlates strongly with unit price (0.80) and not with discount or rating.
- Average customer rating is 3.06 / 5, with about 39% of orders rated 1 or 2.

G.2 Work Summary by Internship Phase

Period	Phase	Work carried out
05.08.2026 – 07.08.2026	Orientation	Understood the problem statement and the data analytics workflow
08.08.2026 – 11.08.2026	Data understanding	Studied dataset fields and structure
12.08.2026 – 14.08.2026	Cleaning	Checked missing values, duplicates, formats and unusual values
15.08.2026 – 17.08.2026	EDA	Computed revenue and descriptive statistics
18.08.2026 – 19.08.2026	Analysis	Product, region, channel and customer summaries; charts
20.08.2026	Final review	Interpreted results and prepared the internship report

G.3 Glossary

Term	Meaning
EDA	Exploratory Data Analysis – examining data using statistics and charts before drawing conclusions
DataFrame	Two-dimensional table structure in Pandas used to hold the dataset
Aggregation	Combining many records into a summary value such as a sum, count or mean
Median / IQR	Middle value of the data / the spread between the 25th and 75th percentiles
Correlation (r)	Measure of linear relationship between two numeric variables, from −1 to +1
Outlier	A value that lies unusually far from most of the data
Revenue	Unit_Price × Quantity × (1 − Discount_Rate) for an order
Segmentation	Grouping customers or orders into age groups or low, medium and high value tiers

INTERNSHIP COMPLETION CERTIFICATE

The internship completion certificate issued by Wiztech Automation Solutions, Chennai, dated 20 August 2026, states that **Mr. R YATHIN YASHWANTH (REG NO: 110725102277)**, studying B.Tech (AI&DS) at J.N.N Institute of Engineering, successfully completed his internship in Data Analytics from 05.08.2026 to 20.08.2026.



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20 AUG 2026

COMPLETION CERTIFICATE

This is to certify that **Mr. R YATHIN YASHWANTH (REG NO: 110725102277)** studying **B.TECH (AI&DS)** at **"JNN INSTITUTE OF ENGINEERING"** has successfully completed his Internship in **DATA ANALYTICS** in **"WIZTECH AUTOMATION SOLUTIONS"** from **05.08.2026** to **20.08.2026.**

We found the Intern to be hard-working, conscientious and responsible during his tenure with us.

We wish him all the best and success.

With Regards,

WIZTECH AUTOMATION SOLUTIONS CHENNAI 600 102

DIRECTOR